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Employee Attendance and Late Arrival Alert System Using Facial Recognition

This project is a high-performance, professional-grade **IoT Facial Attendance System** designed specifically for the Raspberry Pi. This guide covers **every step** from hardware wiring to software deployment.

🏗️ Phase 1: Hardware & Power Connections

Before turning on the software, you must wired the components correctly.

1. Powering the System

- **Main Power**: Use a **5V 3A Micro-USB (or USB-C)** power adapter for the Raspberry Pi.
- **Voltage Rails**:
  - Connect the **VCC** of the LCD and Buzzer to the **5V Pin** (Pin 2 or 4) on the Raspberry Pi.
  - Connect the **GND** of all components to a **Ground Pin** (e.g., Pin 6) on the Pi.

2. Component Wiring (GPIO)

Component	Pin Name	Raspberry Pi Pin
I2C LCD	VCC	5V (Pin 2)
I2C LCD	GND	Ground (Pin 6)
I2C LCD	SDA	GPIO 2 (Pin 3)
I2C LCD	SCL	GPIO 3 (Pin 5)
Buzzer	Positive (+)	GPIO 18 (Pin 12)
Buzzer	Negative (-)	Ground (Pin 14)
Camera	Ribbon/USB	Camera Port or USB Port

📶 Phase 2: Network & Remote Access

The Pi is designed to be headless (no monitor needed).

1. Connect to WiFi

The Pi will automatically search for this network:

- **WiFi SSID (Name):** `admin`
- **WiFi Password:** `123456789`

2. How to Download & Install PuTTY (Windows)

If you don't have PuTTY on your laptop, follow these steps:

1. Go to: www.putty.org
2. Click the link that says **"You can download PuTTY here"**.
3. Under **"Package files"**, look for the **64-bit x86** installer (`.msi` file).
4. Download it and run the installer.
5. Click **Next** through the setup until it is installed.

3. Remote Control (PuTTY)

1. Connect your laptop to the `admin` WiFi.
2. Open **PuTTY**.
3. **Host Name**: `admin2027` (If this fails, use the IP address shown on the LCD).
4. **Login User**: `admin`
5. **Password**: `123456789`

🚀 Phase 3: Software Execution

Step 1: Enter Project Directory

```
cd ~/Employee_attendance
```

Step 2: Sync Latest Code

```
git fetch origin
git reset --hard origin/main
```

Step 3: Start the System

```
python3 run_system.py
```

Wait 5 seconds. The LCD will show the Welcome message and your IP address.

🧠 Phase 4: How the System Works (Full Logic)

1. Facial recognition AI

The system uses a **Hybrid AI Strategy**:

- **Haar Cascade**: Scans 30 frames per second to find "shapes" of faces (low power).
- **Dlib Deep Learning**: Once a face is stable, it extracts a **128-point neural map**. This is unique like a fingerprint.
- **Matching**: It compares your live map against the `.npy` files in `database/faces/`.

2. Shift Timing & Rules (Kolkata IST)

- **Morning Shift (Check-IN)**:
  - **Arrival < 9:30 AM**: Marked as **"On-Time"**.
  - **Arrival 9:31 AM - 12:00 PM**: Marked as **"Late Arrived"**.
- **Evening Shift (Check-OUT)**:
  - **Anytime after IN and before 5:30 PM**: Marked as **"Early Leaving"**.
  - **After 5:30 PM**: Marked as **"Left"** (Normal Checkout).
- **Anti-Spam**: If you try to scan twice after checking out, the LCD says **"OUT Already Done"**.

📊 Phase 5: The Web UI (Dashboard)

Access the dashboard by typing the Pi's IP into your laptop browser (e.g., `http://192.168.1.15:5000`).

- **Dashboard**: Real-time stats of how many workers are in/late.
- **Employee Directory**: Manage staff and verify if their Face ID is registered.

- **Full Logs**: Look at history for any specific date.
- **Data Export**: Click "Download" to get an Excel-ready CSV of all records.
- **Erase**: Completely delete an employee and all their logs from the combined database.

🔊 Signal Reference

- **LCD: "Scanning..."**: AI is currently analyzing your face.
- **Buzzer: 1 Long**: Successful Attendance.
- **Buzzer: 3 Short**: Late or Early scan recorded.
- **Buzzer: 5 Rapid**: Error or Already Scanned.